modified theory.

Yukawa assumes that between the nuclear field and the electrons, positrons and neutrinos there is still another species of particle, which may be called *Yukawa particle* for the time being. These Yukawa particles are assumed to have a mass several hundred times that of an electron, and to be capable of disintegrating into electrons, positrons and neutrinos, directly or eventually through other decay processes. So, according to Yukawa's theory in nuclear transmutations, such a Yukawa particle should actually be emitted. However, this does not happen, because the Yukawa particle has such a great rest mass that the energy, mc^2, necessary for its formation is not available. But the Yukawa particle can break up (directly, or indirectly through other processes) into electrons and neutrinos, and this happens, in a certain sense, in the moment of its formation, so that on the whole, it is sufficient to provide the energy required for the formation of the light particles, the electron and neutrino. This theory thus regards the process of nuclear transmutation as occurring in several steps. First, the Yukawa particle is formed from the nuclear field—or more correctly, the nuclear field itself is identical with the Yukawa particle, which for lack of sufficient energy for its formation cannot manifest itself as a real particle. Instead, no sooner is it formed than it breaks up into electrons and neutrinos, which then actually leave the nucleus.

If we accept this theory as a working hypothesis—and there is much in it to make it plausible—there arises the question whether the Yukawa particles are perhaps identical with a certain species of particle already observed in cosmic radiation. Actually, the most recent experiments make it extremely probable that the role of the Yukawa particles is played—in part, at any rate—by the heavy mesons (or 'π particles') observed by Powell; for in cases of nuclear fission of very high energy, these π particles have been observed to be hurled forth from the nuclei. The π particles (already mentioned on page 55) are about 275 times as heavy as an electron. According to Powell's observations, they first break up into a light meson and a neutral particle (the latter is probably simply a neutrino). This light meson (its mass is about 213 times the mass of an electron) then breaks up further into an electron and probably two neutral particles. It is recognized here that the emission of electrons and neutrinos can occur by very roundabout ways only, possibly due somehow to the fact that the probability of the occurrence of a beta decay is extraordinarily small compared with the probability of other nuclear changes.

These considerations show also that the problem of the relationship between nuclear forces and the elementary particles regarded as being linked with them is a very complex one, which cannot be solved for many more years to come. At this moment, the only thing we know for certain is that the nuclear forces are, to a considerable extent at any rate, exchange forces, and that there exist unstable elementary particles, the mass of which is between